

John Rugemalila

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EDUCATION

The University of Chicago

Expected, May 2027

B.A. in Computer Science

Chicago, IL

- **Relevant coursework:** Data Structures & Algorithms, Distributed Systems, Software Development, Databases

SKILLS

Languages: Java, Python, Go, Rust, TypeScript, JavaScript, C, SQL, GraphQL

Frameworks & Libraries: gRPC, FastAPI, Spring Boot, React, Next.js, PyTorch, scikit-learn, NumPy

Tools & Platforms: AWS, Snowflake, Kafka, PostgreSQL, MongoDB, ChromaDB, Firebase, Linux, Git, Docker

EXPERIENCE

HubSpot

June 2026 – Present

Software Engineer Intern

Boston, MA

- Built a distributed async pipeline connecting ingestion to data privacy/classification, using task queues, idempotent Java workers, retries, and Snowflake processing to eliminate privacy-review gaps across 7 ingestion source types.
- Reduced ingestion complexity and edge-case failures by introducing and backfilling explicit Snowflake object-type metadata, removing manual parsing and special-case logic while adding safeguards for table lifecycle operations
- Improved self-service ingestion for 90+ analytics and engineering teams by simplifying source configuration workflows and surfacing Snowflake table, view, and Iceberg metadata in the internal React platform
- Reduced cross-region Iceberg registration latency from 2+ minutes to sub-second by moving synchronous RPC provisioning to async workers and separating physical Iceberg locations from user-facing Snowflake views

Data for Common Good

June 2025 – Sept 2025

Software Engineer Intern

Chicago, IL

- Developed a GraphQL-based ETL pipeline that consolidated data across 5+ pediatric cancer consortia
- Implemented a caching layer for AWS Lambda-queried datasets, reducing average response times by over 70%
- Optimized GraphQL performance through query consolidation, decreasing API calls by 60%
- Refactored legacy code, improving readability and enabling easier integration across research workflows

Computer Science Instructional Laboratory

March 2024 – May 2026

Software Developer

Chicago, IL

- Developed SemaDoc, an in-house AI-powered documentation tool used across 8 teams, reducing documentation time from hours to minutes using LLM-powered workflows
- Architected the FastAPI backend for SemaDoc, integrating async Trello APIs, Ollama-hosted LLMs, and MongoDB to support 500+ concurrent users with sub-100ms database operations
- Contributed to rewriting the CSIL website using Next.js and Node.js, serving 5,000+ UChicago students
- Built a secure alumni database system with a FastAPI backend, enabling efficient data entry and search

PROJECTS

CrustyDB | [Rust](#)

Jan 2026

- Built the slotted-page layout and heap-file storage manager for CrustyDB, an academic relational database
- Implemented the aggregate, hash join, and nested-loop join operators in its volcano-style execution engine

Raft key-value store | [Go](#), [gRPC](#), [Protocol Buffers](#)

April 2026

- Implemented Raft from scratch, covering leader election, log replication, and commit-index advancement
- Layered a replicated key-value store over the consensus module, using gRPC and Protocol Buffers for the node-to-node interface

spotmv | [github](#) | [Python](#), [Spotify Web API](#)

June 2026

- Built a Spotify CLI that moves an artist's tracks between playlists in bulk, with sorting and renaming
- Made every destructive action a dry run by default, writing changes only after explicit confirmation

http-server | [github](#) | [Go](#)

July 2025

- Built an HTTP/1.1 server from scratch with persistent connections, static file serving, custom routing, and concurrent request handling